S&P 500 quick check – masked-rich panel (single seed), for discussion

Date: 2026-08-22. Status: exploratory single-seed run (NOT the paper's submitted result; the submitted paper is Vietnam-only). Purpose: preview whether the deep/graph models behave on a large market before a full multi-seed S&P 500 study.

Setup

- Design: same masked union-of-dates panel and five node features as the paper's VN Tables 1-2 (`masked-union-panel-rich-5feat-weighted-gat`), lookback = 10, per-node positivity floor 1e-2 x train mean, date-clustered Diebold-Mariano inference.
- Universe: S&P 500 constituents, 442 nodes after the min-train-rows drop, 1624 test dates, 717,768 masked test observations at h1.
- Seeds: **1 (seed 42)** – single-seed exploratory run, `--no-corr` (only the directed volume->Parkinson graph, no symmetric-correlation comparator). GARCH not computed for this panel.
- Models: HAR (classic 3-feature OLS: daily/weekly/monthly), HAR-X (5-feature OLS = the 3 HAR features + market PK + volume z-score; this is what the paper labels "HAR"), LSTM (5-feature, no graph), LSTM+GAT (directed volume->Parkinson Top-5 weighted 2-hop). The paper's submitted tables use the 5-feature model (HAR-X) as "HAR"; both are shown here for completeness.
- Source JSONs: `results/masked_rich_floor1e2/sp500_h{1,5,10,22}/result.json`.

All metrics (single seed)

Bold marks the best value per horizon per metric. Lower is better for MSE/RMSE/MAE/QLIKE; higher for R2.

h	Model	MSE ((x1e-7))	RMSE ((x1e-4))	MAE ((x1e-4))	QLIKE	R2
1	HAR	6.276	7.922	**2.359**	**0.3622**	0.3079
	HAR-X	**6.171**	**7.856**	2.361	0.4078	**0.3195**
	LSTM	6.367	7.979	2.453	0.4110	0.2979
	LSTM+GAT	6.384	7.990	2.466	0.4064	0.2960
5	HAR	8.086	8.992	2.679	0.4349	0.1084
	HAR-X	8.062	8.979	2.691	**0.4303**	0.1110
	LSTM	**7.850**	**8.860**	**2.573**	1.0433	**0.1344**
	LSTM+GAT	7.902	8.890	2.637	0.7907	0.1287
10	HAR	8.937	9.454	2.829	0.4824	0.0165
	HAR-X	8.948	9.459	2.842	**0.4775**	0.0153
	LSTM	8.505	9.222	2.767	0.8276	0.0640
	LSTM+GAT	**8.497**	**9.218**	**2.701**	1.0110	**0.0650**
22	HAR	9.647	9.822	2.991	**0.5993**	-0.0603
	HAR-X	9.623	9.810	2.983	0.6150	-0.0576
	LSTM	9.399	9.695	3.259	0.8515	-0.0330
	LSTM+GAT	**9.172**	**9.577**	**2.952**	1.3198	**-0.0080**

Diebold-Mariano contrasts (date-clustered; p-value, favoured model)

A "better" favour means the first-named model has the lower loss.

h	LSTM vs HAR (QLIKE)	(MAE)	LSTM+GAT vs LSTM (QLIKE)	(MAE)
1	0.000 (HAR)	0.000 (HAR)	0.226 (GAT)	0.002 (LSTM)
5	0.000 (HAR)	0.021 (LSTM)	0.000 (GAT)	0.000 (LSTM)
10	0.017 (HAR)	0.408 (LSTM)	0.035 (LSTM)	0.000 (GAT)
22	0.066 (HAR)	0.024 (HAR)	0.192 (LSTM)	0.000 (GAT)

Reading

- **QLIKE:** HAR / HAR-X have the lowest QLIKE at every horizon. The deep models' QLIKE is inflated at h5/h10/h22 (LSTM 0.83-1.04, LSTM+GAT 0.79-1.32), consistent with a positivity-floor tail-collapse artifact at the S&P 500 variance scale (the 1e-2 floor tuned on the VN panels appears too low here). QLIKE for the deep models on this panel is therefore not yet trustworthy and should be re-checked with a higher / scale-aware floor before drawing conclusions.
- **Point error (MSE / RMSE / MAE / R2):** HAR-X leads at h1; the LSTM and LSTM+GAT lead from h5 onward (LSTM lowest MAE at h5 and significantly lower than HAR, $p=0.021$; LSTM+GAT lowest MSE/RMSE/MAE/R2 at h10 and h22). So on point-error metrics the deep models do improve over HAR at the longer horizons on this large market – the opposite of the VN panels where HAR led point error from h5 onward.
- **Graph (LSTM+GAT vs no-graph LSTM):** mixed. It does not help at h1 (QLIKE ns, MAE worse). It lowers the QLIKE blow-up at h5 ($p<0.001$) but raises it at h10 ($p=0.035$ worse). It gives a significantly lower MAE than the no-graph LSTM at h10 and h22 ($p<0.001$). No consistent QLIKE gain over the no-graph LSTM.

Caveats before using these numbers

1. **Single seed only** – no multi-seed mean or dispersion; a full study needs >=3 seeds.
2. **QLIKE floor artifact** – the deep-model QLIKE values are inflated by the floor; re-run with a scale-aware floor before comparing QLIKE across models on S&P 500.
3. **No GARCH** on this panel yet; –**no-corr** (correlation-graph comparator not run).
4. This is **not** in the submitted paper (Vietnam-only); it is preview material for a possible later cross-market version.